Directed Tree Ascent

A Partizan Combinatorial Game on Edge-Labelled Rooted Binary Trees

IE619 — Combinatorial Game Theory (2025)

Game Definition

Definition (Directed Tree Ascent (DTA))

A finite rooted binary tree with each edge labelled **L** or **R**, directed from parent to child. A token starts at the root.

Left moves the token along any adjacent **L**-edge to a child.

Right moves along any adjacent **R**-edge to a child.

Normal play: the player unable to move loses.

Multiple independent trees are combined using the **disjunctive sum**.

Nine Design Criteria

Criterion	DTA
No target condition	last-to-move wins ✓
Novel	no prior literature ✓
Named	"Directed Tree Ascent" ✓
Scalable	arbitrary depth / branching ✓
Disjunctive sum	independent trees sum freely ✓
One-liner per player	yes ✓
Partizan	distinct **L**/**R** edges ✓
Board feel	path vs. branch strategy tension ✓
Interesting math	$\mathbb{Z}$, switches, $*$, $\uparrow$ ✓

Recursive Value Formula

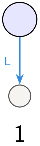
For a node v with **L**-children $\mathcal{L}(v)$ and **R**-children $\mathcal{R}(v)$:

$$\mathcal{G}(v) = \left\{ \mathcal{G}(c) : c \in \mathcal{L}(v) \;\middle|\; \mathcal{G}(c) : c \in \mathcal{R}(v) \right\}$$

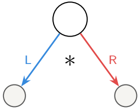
Base case: leaf $\mathcal{G}(\ell) = \{|\} = 0$.

Basic Positions

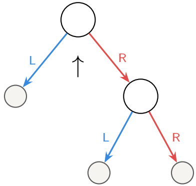
L-path $\to 1$



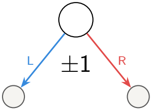
Star $* = \{0|0\}$



Up $\uparrow = \{0|*\}$



Switch ± 1



Value Catalogue

Tree shape	$\mathcal{G}(\cdot)$	Type	$t(\cdot)$
L^n path	n	integer	$-\infty$
R^n path	$-n$	integer	$-\infty$
n L-leaves + n R-leaves	$\pm n$	switch	n
L-leaf + R-leaf	$*$	fuzzy	0
L-leaf + R$\rightarrow$ {L,R}	$\uparrow$	infinitesimal	0
Alternating (odd)	1	integer	$-\infty$

Structural Theorems

Theorem (Monochromatic paths are integers)

$\mathcal{G}(P_L^n) = n$ *and* $\mathcal{G}(P_R^n) = -n$.

Theorem (Symmetric branch gives switch)

Root with n ***L****-leaves and n* ***R****-leaves:* $\mathcal{G}(T_{\pm n}) = \pm n$.

Infinitesimals

Definition

$T_* = \{0 \mid 0\} = *$
$T_\uparrow = \{0 \mid *\} = \uparrow$

Theorem

$\uparrow \| *$ *(they are incomparable).*

Important consequence: Temperature-based strategies can fail when both $*$ and $\uparrow$ appear in a sum.

Outcome Classes

$$\mathcal{G}(T) \begin{cases} > 0 & \Rightarrow \mathcal{L} \quad \text{(Left wins regardless)} \\ < 0 & \Rightarrow \mathcal{R} \quad \text{(Right wins regardless)} \\ = 0 & \Rightarrow \mathcal{P} \quad \text{(second player wins)} \\ \| 0 & \Rightarrow \mathcal{N} \quad \text{(first player wins)} \end{cases}$$

Position	Value	Outcome	Winner
P_L^3	3	$\mathcal{L}$	Left always
$T_{\pm 2}$	± 2	$\mathcal{N}$	First player
T_*	$*$	$\mathcal{N}$	First player
$T_\uparrow$	$\uparrow$	$\mathcal{L}$	Left always
$P_L^2 + P_R^2$	0	$\mathcal{P}$	Second player

Open Problems

(i) Full closed-form value characterisation for arbitrary trees
(ii) Computational complexity of outcome in disjunctive sums
(iii) Misère version of DTA
(iv) k-ary generalisation with k colours
(v) Behaviour on infinite trees

References: *Winning Ways*, *On Numbers and Games*, Siegel's *Combinatorial Game Theory*.

Thank You!

Questions?

Left and **Right** are ready to ascend the tree!